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Unit – 1 : Session – 8 : SLO – 1 | Kernel Data Structures — Symbol Code Puzzle

■ I. Secret Message

- Decoding the symbols with the key at the bottom of the page reveals:
- "KERNEL DATA STRUCTURES KEEP TRACK OF PROCESSES, MEMORY, FILES AND DEVICES."

■ II. What Are Kernel Data Structures?

- PROCESS LIST / PCB TABLE — one Process Control Block per process (state, registers, PC).
- READY QUEUE and WAIT (device) QUEUES — linked lists of process descriptors used by schedulers.
- MEMORY MAPS / PAGE TABLES — track free and allocated frames and address translation.
- OPEN FILE TABLE and INODE structures — track files opened by processes.
- DEVICE TABLE and I/O request queues — track devices and pending operations.